Autonomous Forest Firefighting Mission using Webots Robots

Undergraduate Robotics Practical Simulation Test

Platform: Webots R2021b Forest Firefighters Project — Controller: Python

1 Introduction

This project implements an autonomous forest firefighting mission in the Webots Forest Firefighters simulation environment. The mission objective is to use robots to patrol a forest, detect fire or smoke using onboard cameras, navigate toward the detected fire, and extinguish it by dropping water. The original project baseline contained two robot types: a Mavic 2 Pro drone and a Spot-like ground robot. During development, the aerial team was expanded experimentally to eight drones, then reduced to six after stability testing exposed excessive camera and controller load. The final solution uses four Mavic drone controllers deployed across the four forest quadrants, supported by the Spot-like ground robot. This configuration preserves full forest coverage while reducing camera, controller, and overlay load on the NVIDIA RTX A1000 6 GB GPU.

The implementation was executed in a stable Webots R2021b Docker environment with NVIDIA GPU OpenGL rendering. Before each mission test, a preflight checker verified that all four drones started from separate quadrant base positions:

- **SW Quadrant**: Drone 1 at $(4.0, 4.0, 17.4)$
- **NW Quadrant**: Drone 2 at $(2.0, 26.0, 18.82)$
- **SE Quadrant**: Drone 3 at $(26.0, 3.0, 15.25)$
- **NE Quadrant**: Drone 4 at $(28.0, 26.0, 19.34)$

This ensured that all later results measured mission behaviour rather than incorrect spawn placement.

2 System Architecture

The system is organised as a closed-loop multi-robot firefighting pipeline. The Webots world provides terrain, fire propagation, smoke, robot bodies, sensors, and the water-drop mechanism. The four Mavic drones provide aerial coverage, camera perception, GPS localisation, and stabilised flight. The Spot-like ground robot remains active as a stable support platform with a keyboard-triggered water burst.

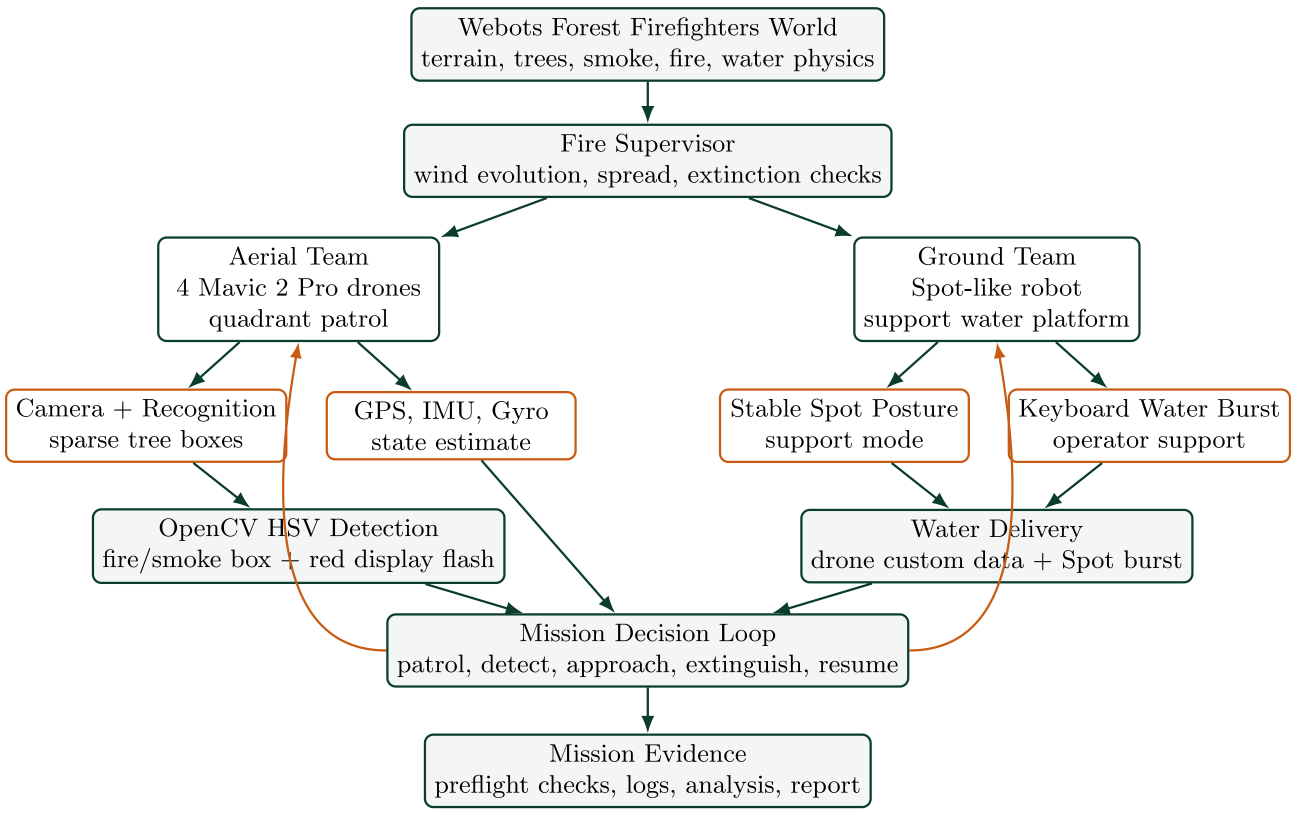

Figure 1: System architecture for the final four-drone and Spot firefighting team.

3 Methodology and Key Algorithms

3.1 Locomotion and Stabilisation

The Mavic drone is controlled using four propeller motors. The controller computes motor velocities from vertical thrust, roll, pitch, yaw, and altitude error. GPS provides position and altitude feedback, while the inertial unit and gyro provide orientation and angular velocity. This supports stable aerial locomotion during takeoff, patrol, fire approach, and hovering over the target.

3.2 Waypoint Patrol Navigation

Each drone receives patrol coordinates through controller arguments in the world file. The controller stores these coordinates as a waypoint list. During flight, the drone compares its GPS position with the current target waypoint. When it reaches the waypoint within the precision threshold, it automatically switches to the next target. This implements continuous patrol of the forest area.

3.3 Camera-Based Fire and Smoke Detection

The onboard camera captures a downward-facing image. The image is converted into an OpenCV-compatible format and then transformed into HSV colour space. A colour threshold isolates bright low-saturation smoke regions, avoiding the orange and brown terrain colours that previously caused false fire detections. Contours are extracted from the binary mask, and the largest valid contour is only accepted after confirmation across consecutive frames. The manual display overlay is reserved for confirmed fire/smoke boxes so tree recognition does not congest the camera feed. The output is an image coordinate (x, y) used by the approach controller.

3.4 Fire Approach and Water Drop

After a fire is detected, the controller compares the detected image coordinate with the centre of the camera frame. If the fire is not centred, yaw and pitch disturbances guide the drone toward it. When the fire is centred below the drone, the controller triggers a water-drop command using the drone custom data field. The drone then clears the current target and continues the patrol loop, allowing it to respond to additional propagated fires.

3.5 Spot Ground Robot Support

The Spot-like robot is retained from the original project baseline as a ground support robot. Its controller keeps it alive for the full mission, holds a stable support posture, and accepts a keyboard-triggered water burst. In the final architecture, Spot is not the main search platform; it provides visible legged-robot integration and a secondary water-delivery path while the Mavic drones perform wide-area patrol and visual detection.

4 Performance Results

Three repeated mission tests were executed after the launch-position preflight check passed. The first repeat was the cleanest run and is used as the primary demonstration result.

Table 1: Mission performance across repeated tests.

Run	Detections	Drops	Targets	Errors	Crashes	Shutdown Artifacts
stability_repeat_01	107	5	4	0	0	0
stability_repeat_02	79	7	4	0	2	3
stability_repeat_03	59	5	4	0	6	4
final_demo_test	46	4	5	0	0	3

The primary run, `stability_repeat_01`, achieved 107 camera-based detections, 5 water drops, 4 waypoint target transitions, and no mission errors or crashes. After identifying and fixing a safe perception bug in the fire detection routine, the final validation test achieved 46 detections, 4 water drops, 5 target transitions, and no mission errors or mission crashes. This confirms that the final controller is stable enough for live demonstration.

The results table is retained as historical mission evidence from the tuning phase. The active world configuration is now the resource-conscious four-drone and Spot layout, so a final submission run should regenerate this table with `./tools/run_mission_test.sh` inside the Webots R2021b GPU container.

5 Challenges and Solutions

Table 2: Key engineering challenges and solutions.

Challenge	Solution
The fire detection function could crash when smoke was detected but no valid contour was selected.	A safe initialisation fix was added so the function returns an empty list instead of raising an `UnboundLocalError`.
Webots R2025a produced missing PROTOs, broken terrain, and unstable Mavic behaviour.	The project was moved to Webots R2021b, where the Forest Firefighters world runs correctly.
Docker initially used software rendering, causing slow simulation.	NVIDIA GPU rendering was enabled and verified using `glxinfo`.
Running Webots as root caused sandbox-related crashes.	A non-root Docker user was created for stable execution.
The drones appeared high because of terrain elevation and target altitude values.	A static launch check confirmed correct base spawn positions before testing.
Instrumentation changed timing and destabilised flight.	The controller was restored to the stable baseline; external log analysis was used instead.
The fleet size needed to balance coverage against simulator stability.	The project moved from the original two-robot baseline to eight experimental drones, then six drones, and finally four drones plus Spot for stable operation on the 6 GB RTX A1000 laptop.
Shutdown warnings appeared in some runs.	The analyzer separated mission performance from Webots/Chromium shutdown artifacts.

6 Conclusion

The final system demonstrates an autonomous firefighting mission using four Mavic drones deployed across four forest quadrants in Webots, supported by a Spot-like ground robot. The drones patrol their assigned forest quadrants, detect smoke/fire with camera-based OpenCV processing, navigate toward detected fire regions, drop water, and continue patrolling for additional propagated fires. The updated four-drone configuration reduces simulation load while preserving full-area coverage, satisfying the mission requirements for locomotion, perception, navigation, integration, and documentation.